

ADHD Across Cultures: Is There Evidence for a Bidimensional Organization of Symptoms?

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We examined the phenotypic expression of attention-deficit/hyperactivity disorder (ADHD) across cultures by assessing the factor/latent class structure of its core symptoms. We conducted a systematic review of the literature published from January 1987 to November 2008 using Medline and PsycINFO. We systematically reviewed 2,511 article abstracts, and 48 of these abstracts were included in this review. Research with school-age children from 15 countries including different samples, informants, and rating instruments supported a two-factor ADHD model consisting of inattention and combined hyperactivity and impulsivity. This model was not supported for preschool children. Cross-cultural equivalence for the two-factor model was suggested in some studies. Latent class analyses using parental data and more recent approaches like factor mixture modeling are generally consistent with factor analyses. These findings argue in favor of a cross-cultural validity of the syndrome. The implications of these findings for further research and classificatory systems in mental health like the *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.) are discussed.

There is a strong debate in the literature regarding whether psychiatric disorders are universally the same or differ on their core definitions and constellation of symptoms, as influenced by cultural factors (Canino & Alegria, 2008). This issue has been particularly controversial for attention-deficit/hyperactivity disorder (ADHD). ADHD is a highly prevalent developmental disorder characterized by a persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with appropriate social, academic, or occupational functioning (American Psychiatric Association [APA], 1994).

As early as in the 1980s, scholars suggested that hyperactivity was a Western concept that had no validity in other cultures. The disorder was associated with “cultural tempo” (Block, 1977) or “permissiveness” of Western culture (Ho, 1981). A medullar argument used by the proponents of these views is that ADHD would be present in some cultures but not in others because culture shapes the individual’s development and his or her biological and psychological unfolding to a substantial degree. The controversy that originated from these views continues in the clinical literature (Ali, 1996; Anderson, 1996) with an emphasis on the conceptualization of ADHD in terms of the influence of Western sociocultural factors (Timimi & Taylor, 2004).

The extent to which ADHD is a relevant construct across cultures is a matter to be determined by empirical inquiry that establishes the validity of the diagnostic

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criteria worldwide (Bird, 2002; Canino & Alegria, 2008). This clarification is conceptually important since the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-IV; APA, 1994)* attempts to provide descriptions of disorders that can be applied across cultures, ethnicities, and contexts. Evidence of possible variations in the symptom expression of ADHD across cultures and ethnicities can inform workgroups in charge of developing the fifth edition of the *DSM (DSM-V)* about revisions of ADHD criteria as defined in the previous edition.

In this article, we review the literature examining whether data from research across different cultures identify similarly the organization of ADHD symptoms. Specifically, we examine whether the core symptoms of ADHD tend to demonstrate comparable structures when assessed in different geographical areas or cultures, by reviewing the research literature on factor analysis of ADHD symptoms. Similarities in the symptom expression of ADHD would support the face validity of the disorder, and is the first of five criteria modified by Bird (2002) to determine the validity of cross-cultural psychiatric diagnoses (Robins & Guze, 1970). Given the scope of the proposed work, we examine only this criterion. The other criteria (i.e., commonality in risk factors, comorbidity, outcomes including neurobiological tests, and treatment response) need to be examined in separate literature reviews.

Three models have been advanced to describe the structure of ADHD symptoms, each one corresponding to descriptions of the disorder as proposed in *DSM* taxonomies (APA, 1980, 1987, 1994). The unidimensional model implied in the revised third edition of the *DSM (DSM-III-R; APA, 1987)* taxonomy posits that symptoms of inattention, hyperactivity, and impulsivity are organized into a single factor of maladaptive behaviors. The two-factor or bidimensional model implied in the *DSM-IV* taxonomy assumes that symptoms organize into two factors: independent inattention and combined hyperactivity-impulsivity (APA, 1994). Finally, the three-dimensional model implied in the third edition of the *DSM (DSM-III; APA, 1980)* taxonomy proposes that ADHD presents symptoms in domains of inattention, hyperactivity, and impulsivity (Lahey et al., 1988). In addition to reviewing the factor analytic literature, we examine recent investigations using latent class analyses to confirm if their findings concur with studies that used exploratory or confirmatory factor analyses.

As pointed out by Wolraich et al. (2003), detecting similar factor dimensions in different geographical regions or cultural samples would support the view that the organization of symptoms of ADHD reflect children's behavior as opposed to the value laden culturally distinct perception of parents, teachers, or clinicians.

It would also help in the conceptualization and definition of the disorder. Evidence of comparable factor dimensions similarly associated with gender and age effects would further support a cross-cultural consistency of the underlying dimensions of ADHD.

METHOD

Review Procedures

We conducted a systematic review of the literature published from January 1, 1987, to November 17, 2008, using Medline (accessed by PubMed) and PsycINFO, for articles in English, Spanish, or Portuguese. References were searched using the following terms: attention-deficit/hyperactivity disorder, hyperactivity, inattention, factor and/or latent class analyses, preschool, childhood, and adolescents. This review strategy identified 2,511 abstracts, which were reviewed by the first author. We checked relevant references cited in the articles. Forty-eight studies based on *DSM-III-R* or *DSM-IV* were selected for further review if they reported factor and/or latent class analyses findings of ADHD symptoms based on school, community, or clinical samples. If longitudinal data were presented in the studies selected, only the first wave results were considered.

Analytic Strategies of Reviewed Studies on Structural Organization of ADHD Symptoms

Researchers have used factor analysis to examine the underlying dimensions of ADHD. This factor analytic strategy measures which of several observable variables or symptoms correlate with an assumed underlying dimensional structure. In exploratory factor analyses (EFA), theoretical considerations about factor composition may or not be considered and the correlations between variables determine what factors will be identified. In confirmatory factor analysis (CFA), however, factor models are specified based on theory. The fit of each model is computed and the models are then compared, providing information on which of the concurrent models best represents the data (Kim & Mueller, 1978).

A critical test of the equivalence across groups of the structural organization of ADHD symptoms requires that ratings across participants be analyzed simultaneously using a multiple-group CFA or other statistical approaches (Gomez & Vance, 2008). The multiple-group CFA strategy can test whether (a) the different groups associate the same items with the same latent construct, that is, have the same factor structure (configural invariance); (b) the strength of the relationships between the same items and their underlying constructs are the same across the different groups, that is, same

factor loadings (metric invariance); (c) the same items in the different groups have the same amount of unique invariance (error variances invariance); and (d) the same raw score endorsed by the groups is related to the same latent score for the groups (scalar invariance; Gomez & Vance, 2008). The test for scalar invariance is considered essential for demonstrating cross-cultural equivalence as it indicates if there are similarities across groups in the way items are interpreted (Cheung & Rensvold, 2000; Gomez & Vance, 2008).

There are two different perspectives in the literature describing the organization and distribution of ADHD symptoms in a given population. The first describes ADHD as dimensional trait and the second as a categorical or discrete entity (Rohde, 2008). Latent class analysis (LCA) is a statistical procedure that can be used to disentangle this issue as it assumes that observable variables are imperfect indicators of an underlying variable with a finite number of mutually exclusive levels or classes. Thus, LCA can be conceptualized as a categorical variant of factor analysis, whereas EFA/CFA assumes a model of continuous distributed latent variables (Hudziak et al., 1998; Neuman et al., 1999).

RESULTS

Studies Reviewed

A large number of studies have examined the factor structure of ADHD symptoms as defined in the *DSM-III-R* and *DSM-IV* taxonomies. As shown in Table 1, these factor analysis studies vary depending on the country, type of sample (school, community or clinical), age range of the children included (3–18 years), analytical strategy used (EFA, CFA), and informants (teacher, parent, child, clinician). ADHD symptoms were assessed using diagnostic interview schedules designed to measure psychiatric disorders, behavior rating scales, or specific *DSM*-based rating scales developed for measuring symptoms but not disorders. The reviewed studies in Table 1 can be classified into three broad categories: (a) those that examine models for the organization of ADHD symptoms; (b) those that examine models for the organization of ADHD, oppositional defiant disorder (ODD), and/or conduct disorder (CD) symptoms; and (c) those that simultaneously compare the organization of ADHD symptoms in samples from different countries, languages, and cultures. These studies are summarized next.

Organization of ADHD Symptoms

Research that used EFA as an analytic strategy consistently report findings that support the bidimensional

model of ADHD with relatively independent inattention and hyperactivity-impulsivity symptom dimensions. These findings were based on teachers' and/or parents' ratings of children from the United States (Healey et al., 1993; Hudziak et al., 1998), Brazil (Brito, Pinto, & Lins, 1995), Puerto Rico (Bauermeister, Alegria, Bird, Rubio-Stipec, & Canino, 1992; Bauermeister et al., 1995), Finland (Lubke et al., 2007), and South Africa (Meyer, Eilertsen, Sundet, Tshifularo, & Sagvolden, 2004). The results were also based on self-ratings of adolescent from the United States and Brazil (Rasmussen, Todd, et al., 2002; Rohde et al., 2001), and clinicians' ratings of ADHD symptoms based on interviews of parents in a sample of children from 11 European countries and Australia, Israel, and South Africa (Zhang, Faries, Vowles, & Michelson, 2005). EFAs of clinicians' ratings of clinic-referred children also replicated a two-factor model in each of the 7 European country-specific (e.g., France) samples or groups of country (e.g., Nordic countries such as Denmark–Iceland–Norway) samples (Dopfner et al., 2006). The only exception to the findings summarized in this section was reported by Hurtig et al. (2007) for a group of Finnish adolescent ages 16 to 18 years with a diagnosis of ADHD. In this community study, four factors were identified: distractibility, dreamy, hyperactivity, and impulsivity.

Research that used CFA as an analytic strategy were generally supportive of the two-factor model. Gomez, Burns, Walsh, and de Moura (2003) examined the structural organization of parent and teacher ratings of ADHD symptoms in Australian and Brazilian children. In both national samples, the two-factor model provided a good fit in the absolute sense and a better fit than the one-dimensional factor model. DuPaul et al.'s (1997) findings also supported the bidimensional model for teachers' ratings of U.S. children ages 4 to 19. Amador-Campos et al.'s (2005) analyses of teachers' and parents' ratings of Spaniard children ages 4 to 12 supported both a two-factor and a three-factor model, with the bidimensional model considered more parsimonious. For both parents' and teachers' ratings of Australian children ages 5 to 11, Gomez, Harvey, Quick, Scharer, and Harris (1999) reported that the two- and three-factor models were supported by the data across and between gender groups. The authors indicated that the fit of the three-factor model was only slightly better but that their findings supported the two-factor model as an appropriate model for the organization of the ADHD symptoms. Scholte, van Berckelaer-Onnes, and van der Ploeg's (2001) analysis of clinicians' ratings of Dutch children ages 4 to 18 in a residential treatment center supported a three-factor model. Finally, Muris and Meesters (2003)

TABLE 1
Summary of Worldwide Studies That Tested the Factor Structure of ADHD Symptoms

<i>First Author and Country</i>	<i>Sample</i>	<i>Age/Grade</i>	<i>Method</i>	<i>DSM</i>	<i>Source</i>	<i>Dimensions</i>
I. Study that examined organization of ADHD symptoms						
Amador (2005) Spain	School 1,018	4–12	CFA	IV	T,P	2-factor and 3-factor models; 2-factor model selected
Bauermeister (1992) Puerto Rico	Community 614	6–16	EFA	III-R	T	2-factor model
Bauermeister (1995) Puerto Rico	Community 248	9–17	EFA	III-R	T,P	2-factor model
Brito (1995) Brazil	School 2,082	M = 11.2	EFA	III-R	T	2-factor model
Dopfner (2006) Netherlands; UK; Austria-Switzerland; France; Italy; Denmark-Iceland-Norway; Germany	Clinic 1,478	6–18	EFA	IV	CI	2-factor model was replicated in all samples. 3-factor model was extracted in Austria-Switzerland, France, Germany, and UK samples.
DuPaul (1991) U.S.	School 653-P; 564-T	6–12	EFA	III-R	T,P	2-factor model
DuPaul (1997) U.S.	School 4,130	4–19	EFA & CFA	IV	T	2-factor model for EFA and CFA
Gomez (1999) Australia	School 1,275	5–11	CFA	IV	T,P	The fit of the 3-factor model was only slightly better than the fit of the 2-factor model but the latter was construed as more appropriate
Gomez (2003) Australia & Brazil	School Brazil 285; Australian 1,475	M = 8.91 M = 8.28	CFA	IV	T,P	2-factor model for Australian and Brazilian parents and teachers, respectively.
Hardy (2007) U.S.	Clinical 532	3–5.5	CFA	IV	T,P	For parents, 2-factor and 3-factor models were marginally acceptable. For teachers, none of the models was acceptable.
Healey (1993) U.S.	School 85	6–12	EFA	III, III-R	T	2-factor model for both DSM-III and DSM-III-R
Hudziak (1998) U.S.	Female twins 1,546	13.5–19.5	EFA	IV	P	2-factor model
Hurtig (2007) Finland	Community 163 ADHD dx	16–18	EFA	IV	CI	4-factor model: Distractibility, Dreamy, Hyperactivity, and Impulsivity
Lubke (2002) Finland	Community 6622	16–18	EFA	IV	P	2-factor model
Meyer (2004) South Africa	School 6,094	6–15	EFA	IV	T	2-factor model for each of 6 language groups
Muris (2003) Netherlands	School 558	10–14	CFA	IV	P,C	1-factor, 2-factor, and 3-factor models fitted the data well. The 1-factor model was more parsimonious
Rasmussen (2002a) U.S.	Community boys 497	12–14	EFA	IV	C	2-factor model
Rhode (2001) Brazil	School 1,013	12–14	EFA	IV	C	2-factor model
Scholte (2001) Netherlands	RTC 412	4–18	CFA	IV	CI	3-factor model and 2-factor models, slightly favoring the 3-factor model
Zhang (2005) Australia, Israel, South Africa, & 11 European countries	Clinical 604	6–15	EFA	IV	CI	2-factor model
II. Studies that examined organization of ADHD and ODD/CD symptoms						
Bauermeister (1992) Puerto Rico	Preschool 665	4–5	EFA	III-R	T	Pre-school: 2-factor model (ADHD & ODD)
	School 680	6–13				Elementary school: 3-factor model (I and HI) and ODD
Baumgaertel (1995) Germany	School 1,077	Grades 1st–4th	EFA	III-R, IV	T	4-factor model: I, HI, ODD, CD
Burns (1997) U.S. (two studies)	School 1,445 & 1,711	M = 8.58, 9.12	CFA	III-R, IV	T	4-factor model: I, HI, ODD, CD in both studies

(Continued)

TABLE 1
Continued

<i>First Author and Country</i>	<i>Sample</i>	<i>Age/Grade</i>	<i>Method</i>	<i>DSM</i>	<i>Source</i>	<i>Dimensions</i>
Burns (2001) U.S.	Pediatric unit 1,015	3–16	CFA	IV	P	3-factor model (I and HI) and ODD
Gomez (2005) Malaysia	School 917	Boys M = 8.87 Girls M = 9.02	CFA	IV	T,P	3-factor model (I and HI) and ODD
Hartman (2001) Netherlands, Canada, & U.S.	Community & clinic 12,892	Overall: 3–19	CFA	IV	T,P	I, HI, CD, ODD, generalized anxiety & depression factors were supported by the data
Molina (2001) U.S.	School 247 Clinic 247	11–16 School 13–18 Clinic	CFA	IV	T	3-factor model (I and HI) and ODD in both samples
Pelham (1992) U.S. & Canada	School boys 931	5–14	EFA	III-R	T	3-factor model (I and HI) and ODD-CD symptoms
Pillow (1998) U.S.	Clinic 300	5–15	CFA	IV	P	3-factor model (I and HI) and ODD-CD symptoms
Zuddas (2006) Italy	Teacher 1,085 Parent 1,575	6–12	CFA	IV	T, P	4-factor model (I and HI), ODD, and CD
III. Studies that simultaneously compared organization of ADHD symptoms in different groups						
Beiser (2000) U.S. & Canada	Native 1,109 Non-Native 401	Grades 2nd & 4th	MG CFA	III-R, IV	T,P,C	2-factor model for parents and teachers in native and non-native samples; 1-factor model for children
Gomez (2009) Australia & Malaysian	School Australian 783 Malaysian 928	6–12	MG CFA	IV	P	2-factor model for Australian and Malaysian samples
Reid et al. (1998) U.S.	School boys Caucasian 1,359 African American 381	5–18	SEM	IV	T	2-factor model was appropriate for Caucasian and African American samples
Wolraich (2003) U.S. Suburban U.S. Urban Germany Spain	11,092 6,090 1,077 1,283	Grades K-5th K-4th 1st–4th 1st–4th	MG CFA	IV	T	2-factor model showed the best fit for all samples; a 3-factor model was detected but had negligible importance in the analyses

Note. ADHD = attention-deficit/hyperactivity disorder; DSM = *Diagnostic and Statistical Manual*; CFA = confirmatory factor analysis; EFA = exploratory factor analysis; T = teachers; P = parents; C = child; Cl = clinician; 1-factor model = all ADHD symptoms; 2-factor model = Inattention (I) and Hyperactivity-Impulsivity (HI); 3-factor model = Inattention, Hyperactivity and Impulsivity; M = male; F = female; RTC = residential treatment center; MG CFA = multigroup CFA; SEM = structural equation model.

found that each of the three models fit the data well for Dutch parents' ratings and children's self-ratings of ADHD/Hyperkinetic Disorder symptoms. However, the authors indicated that the one-factor model was more parsimonious. Of note are the findings reported for clinic-referred preschoolers in the United States. For parents' ratings, the two- and three-factor models were marginally acceptable, whereas for teachers' ratings, none of the models was acceptable (Hardy et al., 2007). In brief, the majority of findings from CFA studies of school-age children supported a two- or three-dimensional organization of ADHD symptoms.

Organization of ADHD, ODD, and/or CD Symptoms

Given the high rates of comorbidity between ADHD, ODD, and CD, it has been argued that a more comprehensive and clinically relevant model of assessment

needs to include the symptoms of these three disorders (Pillow, Pelham, Hoza, Molina, & Stultz, 1998). Both EFA and CFA of these sets of symptoms consistently yielded data (Table 1) that supported a three-factor model (inattention, hyperactivity-impulsivity, ODD or ODD-CD behaviors) or four-factor model (inattention, hyperactivity-impulsivity, ODD, and CD behaviors). These models were supported from analyses of teachers' or parents' ratings of clinic referred and non-clinic-referred children in the United States (Burns et al., 2001; Burns, Walsh, Owen, & Snell, 1997; Molina, Smith, & Pelham, 2001; Pelham, Gnagy, Greenslade, & Milich, 1992). They were also supported from analyses of parents' and teachers' ratings of Italian children (Zuddas et al., 2006) and teachers' ratings of German children (Baumgaertel, Wolraich, & Dietrich, 1995). Further support was provided by parents' and teachers' ratings of Malaysian children (Gomez, Burns, Walsh, &

Hafetz, 2005) and teachers' ratings of Puerto Rican primary school children (Bauermeister, 1992). Of interest, in the latter study the emerging model for preschoolers was a two-factor model consisting of ADHD and ODD symptoms. Finally, Hartman et al. (2001) conducted a comprehensive study to evaluate the internal construct validity of a *DSM-IV* based model for ADHD and other syndromes with population and clinic-referred children from the Netherlands, United States, and Canada. The data supported separate dimensions of symptoms of inattention, hyperactivity-impulsivity, ODD, CD, generalized anxiety, and depression. However, the model did not meet an absolute standard of a good model fit due, among other reasons, to the presence of ambiguous symptoms.

Simultaneous Comparisons of the Organization of ADHD Symptoms in Culturally Different Samples

The studies previously reviewed provided evidence of a comparable organizational structure of ADHD symptoms across samples. Rather than analyze and compare the groups concurrently, however, the models were run separately (Gomez & Vance, 2008). Consequently, this evidence does not necessarily address the cultural equivalence or invariance of the two-factor model.

Four studies used multiple-group CFA to examine racial or national configural invariance across groups. Reid et al. (1998) compared teachers' ratings of two groups of Caucasian and African American boys, whereas Gomez (2009) compared parents' ratings of ADHD symptoms in Australian and Malaysian children. In each of these studies, the two-factor model was structurally equivalent in both groups. Wolraich et al. (2003) also evaluated the factor structure of teachers' ratings of ADHD symptoms in a large group of children from Spain, Germany, urban United States, and suburban United States. The analyses suggested that the two-factor model was structurally equivalent for all samples. The authors also reported that a three-factor model was detected but had negligible importance in the analyses. Beiser, Dion, and Gotowiec (2000) examined the relative adequacy of the one- and two-factor models of ADHD symptoms with Native American and non-native American children in different locations across the United States and Canada. For the ratings obtained from parents and teachers, a two-factor solution provided the best fit. The one-factor solution was supported for children self-ratings. In brief, all four studies supported configural invariance for parents' and /or teachers' ratings and, as noted by Gomez and Vance (2008), metric invariance as well. Of interest, in a different study, Gomez and Vance reported that with few exceptions, Malay and Chinese parents in

Malaysia endorsed the same raw scores or ratings for the same levels of the underlying inattention and hyperactivity-impulsivity symptom dimensions (scalar invariance). These findings were interpreted as providing acceptable support for cross-cultural equivalence for the ADHD symptoms across these two ethnocultural groups and the use of the same cut-off scores for the diagnosis of this disorder.

Gender and Age Effects

Nine studies reported higher parent and/or teacher ratings of inattention and hyperactivity-impulsivity symptoms for boys. These studies were conducted in Spain (Amador-Campos et al., 2005), Brazil (Brito et al., 1995), the United States (DuPaul, 1991; DuPaul et al., 1997; Molina et al., 2001), Australia (Gomez et al., 1999), South Africa (Meyer et al., 2004), Puerto Rico (Bauermeister, 1992), and Italy (Zuddas et al., 2006). Gender differences were not found in the two studies where Brazilian adolescents provided self-ratings of ADHD symptoms (Rohde et al., 2001) or when clinicians rated these behaviors in children from several European samples based on parental interviews (Dopfner et al., 2006).

Age effects for the dimensions of inattention and hyperactivity-impulsivity identified in the two-factor model supported by the data are difficult to compare given that the age range of the children rated varied considerably across studies. With this limitation in mind, the cross-sectional data suggested no age effects (DuPaul, 1991; Gomez et al., 1999) or higher parent, teacher, and clinician ratings for older as compared to younger school-age children for the inattention dimension (Brito et al., 1995; Dopfner et al., 2006; Zuddas et al., 2006). In contrast, for the hyperactivity-impulsivity dimension, most studies indicated lower ratings for older children (Brito et al., 1995; Dopfner et al., 2006; DuPaul et al., 1997; Gomez et al., 1999; Zuddas et al., 2006, for parent reports only). The exceptions for the findings just summarized were those reported by Zuddas et al., who reported no age effects for teacher ratings for hyperactivity-impulsivity symptoms, and by Du Paul, who reported no age effects for both symptom dimensions.

LCA

Our literature review was able to identify 14 studies that used LCA to describe the structure of ADHD symptoms in a given population (Acosta et al., 2008; Althoff et al., 2006; Hudziak et al., 1998; Lubke et al., 2007; Neuman et al., 2001; Neuman et al., 2005; Neuman et al., 1999; Rasmussen, Neuman, et al., 2002, 2004; Rasmussen, Todd, et al., 2002; Rohde

et al., 2001; Todd et al., 2001; Volk, Henderson, Neuman, & Todd, 2006; Volk, Neuman, & Todd, 2005). As reported for factor analyses, the integration of findings is extremely difficult because investigations were conducted in samples from different age ranges, proportion of boys/girls and nationalities, using different information sources, and approaches related to the inclusion of comorbidities in analyses.

Basically, in samples using parent data (the majority of studies), the best fit ranged from two to eight classes. The most found classes in different populations tended to be (a) a class with few symptoms endorsement, (b) a class with severe inattentive symptoms endorsement, and (c) a class with severe inattentive and hyperactivity-impulsivity symptoms endorsement (Acosta et al., 2008; Hudziak et al., 1998; Neuman et al., 2001; Neuman et al., 1999; Todd et al., 2001; Volk et al., 2005).

Comorbidity did not have a great impact on the number of classes, type of symptoms included, and level of endorsement in each class in different studies (Acosta et al., 2008; Neuman et al., 2001; Volk et al., 2006; Volk et al., 2005). The same applied for different information source (Althoff et al., 2006). However, a different class solution was determined in samples from the United States and from Brazil in the study by Rasmussen Todd, et al. (2002). The authors hypothesized that this might be related to the collection of data from adolescents in Brazil and from parents in the United States. Regarding cross-cultural comparisons, Rasmussen, Neuman, et al. (2002) documented a similar latent class structure in populations from the United States and Australia using parent reports. Recently, Lubke et al. (2007) used factor mixture analyses that combine factor and latent class approaches in a Finnish Birth Cohort to assess ADHD symptoms. Factor mixture analyses presented a better goodness-of-fit than both factor analyses and LCA. The best solution was a model with two factors (inattentive and hyperactive-impulsive) and two classes based on severity (low endorsement of symptoms majority class and high endorsement of symptoms minority class).

DISCUSSION

We evaluated the face validity of ADHD across cultures by reviewing scientific evidence on the extent to which the organization of the ADHD syndrome is similarly recognized by parents, teachers, clinicians, and children of different geographic regions, ethnicities, and cultures. Overall, our review of the factor analyses of the core ADHD symptoms worldwide supports a cross-cultural consistency of the expression of the underlying dimensions of this disorder. Most studies that examined models for the organization of ADHD symptoms alone or in combination with symptoms of other disorders in

school-age children support a model of inattention and combined hyperactivity and impulsivity symptoms. This bidimensional organization was identified in studies from at least 15 countries in diverse geographic locations such as Africa, Arabian Peninsula, Asia, Europe, North America, Oceania, and South America that used different informants (teacher, parent, child, and clinician) and diverse rating instruments. Furthermore, four of these studies provided evidence of the cultural equivalence of the ADHD bidimensional model across samples by analyzing their factor structure simultaneously through multigroup CFA (Gomez & Vance, 2008). The structural equivalence of the bidimensional model was found in teacher ratings of Caucasian and African American samples in the United States (Reid et al., 1998) and in teacher ratings of Spaniard, German, and urban and suburban children from the United States (Wolraich et al., 2003). It was also found in teacher and parent ratings of Native American and non-native American children from the United States and Canada (Beiser et al., 2000) and in parent ratings of Australian and Malaysian children (Gomez, 2009). These four studies also provide evidence that the strength of the relationships between the same items and their underlying construct (factor loadings or metric invariance) were the same across the different groups. It is important to interpret these findings taking into consideration the fact that the reduced number of impulsivity symptoms (three) has the potential to influence the factor structure of *DSM-IV* symptoms. We cannot rule out the possibility that a larger number of impulsivity items would have yielded a three-dimensional factor structure consisting of symptoms of inattention, hyperactivity, and impulsivity.

A critical evaluation of cross-cultural equivalence is the test for scalar invariance. This test would indicate if there are similarities across groups in the way items are interpreted, that is, the same raw score endorsed by the groups is related to the same latent scores for the groups (Gomez & Vance, 2008). Gomez's (2009) comparison of Australian and Malaysian children yielded partial evidence for scalar invariance. In a different study, Gomez and Vance found support for scalar invariance of Malay and Chinese children in Malaysia. The authors interpreted these findings as providing acceptable support for cross-cultural equivalence for the ADHD symptoms across these ethnic-cultural groups (Gomez & Vance, 2008). Future research needs to examine scalar invariance of ADHD symptoms across other cultural groups.

Additional evidence of cross-cultural consistency in the expression of ADHD symptoms is found in the review of gender and age effects. Findings supported the presence of higher ratings of inattention and hyperactivity-impulsivity symptom dimensions for boys

than girls. The reviewed cross-sectional studies generally support a differential age effect for ADHD symptom dimensions in different cultural groups. For the inattention dimension, no age effects or higher scores for older children were reported. The opposite trend was generally found for symptoms in the hyperactivity-impulsivity dimension. Longitudinal analyses are needed to confirm the validity of these findings.

Although data from LCA are not conceptually comparable with those derived from EFA/CFA (see earlier), the most frequent classes found in different populations tended to be (a) a class with few symptoms endorsement, (b) a class with severe inattentive symptoms endorsement, and (c) a class with severe inattentive and hyperactivity-impulsivity symptoms endorsement. Thus, along the dimensions of inattention and hyperactivity-impulsivity, ADHD symptoms tend to organize in two classes of high endorsement of symptoms, one resembling ADHD inattentive type with low hyperactivity-impulsivity and another with combined symptoms. More important, data from factor mixture analyses, a strategy that combines factor and latent class approaches, documented the same two dimensional solution as those found in the majority of EFA/CFA (Lubke et al., 2007).

It can be argued that some cultures may have alternative ways of expressing ADHD not captured by behavior checklists or the *DSM* symptom lists employed in the studies reviewed. By using these sets of symptoms, investigators may have imposed an appearance of cross-cultural bidimensional consistency that would not have appeared if cultural variations in symptoms had been allowed. As pointed out by Meyer et al. (2004), however, there may be basic similarities in social processes affecting behavior and perceptions of behavior across cultures. As an example, the authors discussed the idea that similar school settings lead to similar expectations and demands, and cause similar definitions of behaviors. Similarities in social processes might cause either fundamentally similar behavior or similar classification of dissimilar behavior descriptions. In the latter case, even if cultural differences are associated with differences in behaviors, and the various response units are different, these could still be perceived as belonging to the same behavior patterns, for example, inattention and hyperactivity (Meyer et al., 2004).

Our survey of the literature argues against the view that ADHD is a Western concept associated with particular sociocultural influences and no validity in other countries, or is a product of the cultural values in particular geographical regions. This conclusion is further supported by the comprehensive review of the worldwide prevalence of ADHD conducted by Polanczyk, de Lima, Horta, Biederman, and Rohde (2007). In other words, our findings suggest that there is more

commonality than differences in the cross-cultural symptom expression of ADHD. It is important to emphasize that our conclusion by no means denies the crucial role that culture and contextual factors play in the expression of ADHD. Culture and the environment can provide important risk and protective factors for the expression of ADHD and the meaning given to these behaviors. Moreover, it can shape the threshold or tolerance of what is considered pathological versus normal behavior, can influence the definition of associated impairments, and can affect the decision to seek treatment as well as the course and response to interventions (Canino & Alegria, 2008).

Implications for Research, Policy, and Practice

The bidimensional model of ADHD symptoms is generally confirmed across school-age children from different geographical regions, cultures, ethnicities, gender, and age groups. It would appear that in spite of cultural influences in the perception and interpretation of behavior, parents, teachers, and clinicians can discriminate between different ADHD symptom patterns in a similar way as implied in the *DSM-IV* (Meyer et al., 2004). These findings, and emerging evidence that ADHD symptoms in the *DSM-IV* are invariant across some cultures (Gomez, 2009; Gomez & Vance, 2008), support retaining this symptom organization model in classificatory systems such as the *DSM-V*.

Although our review support the internal validity of the ADHD symptom dimensions, there is a need to evaluate its discriminant validity across different cultures. Research indicates that the inattention dimension appears to be more strongly associated with executive function deficits (Chabildas, Pennington, & Willcutt, 2001; Nigg et al., 2005), and academic underachievement, shyness, and social withdrawal (Lahey et al., 1998). The hyperactivity-impulsivity dimension, on the other hand, is more strongly associated with accidental injuries and peer relationship problems associated with disruption, aggression, and low self-control; conduct problems; and antisocial behavior (Lahey et al., 1998). Replication of these findings cross-culturally would further support the external validity and generalizability of these two ADHD dimensions.

It is important to note that factor analyses of parent and/or teacher ratings of ADHD symptoms in preschool children do not support a bidimensional model. Future research needs to clarify whether these findings reflect the need to include developmentally appropriate ADHD symptoms for preschoolers in the classification systems, the relative instability of the ADHD syndrome in this age group, or home and/or school environments that do not place significant demands on the attention processes of preschool children.

The cross-cultural stability of the structural organization of the inattention and hyperactivity-impulsivity symptom dimensions in school-age children suggests that dimensional measures of ADHD can be recommended for the *DSM-V* and other classification systems so as to increase the reliability and precision of the evaluation of the severity of this disorder. This recommendation is further supported by the high internal consistency and test-retest reliability of these two symptom dimensions across cultures (e.g., Dopfner et al., 2006; Gomez et al., 1999; Zhang et al., 2005; Zuddas et al., 2006). Finally, levels of inattention or hyperactivity-impulsivity could also be used to qualify or describe the clinical presentation of ADHD. It has been suggested that this approach may be more clinically useful than the current subtyping of ADHD on the basis of inattention and/or hyperactivity-impulsivity symptoms (Lahey, Pelham, Loney, Lee, & Willcutt, 2005; Toplak et al., in press).

However, it is necessary to give serious consideration to establishing severity or diagnostic cut-off scores linked to impairment that take into consideration the normative behaviors of each culture or geographic region, gender, and developmental factors. Responses on rating scales or diagnostic interviews may have different meanings across ethnic or cultural groups (Bauermeister, Berrios, Jiménez, & Acevedo, 1990; Reid, 1995). Also our review shows that boys tend to present higher parent and/or teacher ratings of inattention and hyperactivity-impulsivity symptoms than girls and that symptoms in the latter dimension tend to decrease with age. Emerging cross-cultural literature on children sampled from the community suggests that patterns of associations of ADHD with correlates, comorbidity, and impairment are not different for boys and girls (Bauermeister et al., 2007; Biederman et al., 2005; Graetz, Sawyer, & Baghurst, 2005). Given the reported gender differences in number of ADHD symptoms in our review, there may be girls who have symptoms and impairment that are consistent with ADHD, but they are not currently being identified by *DSM-IV* criteria (Waschbusch & King, 2006). Similarly, findings that symptoms decrease with age may be reflecting a true decline in hyperactivity-impulsivity with age in which case age-specific diagnostic thresholds may be needed. Alternatively, this "decline" may reflect the use of symptoms that are not appropriate to the child's age and or developmental level (Barkley, 2006). These issues merit further research.

Our review support the first of four criteria established for determining the cross-cultural validity of psychiatric disorders (i.e., similar description of ADHD across cultures; Bird, 2002; Robins & Guze, 1970). Literature reviews are now needed on the other criteria related to commonality in risk factors, comorbidity,

outcomes including neurobiological tests, and treatment response. This initiative can lend further support to the cross-cultural validity of ADHD.

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